

Super Surface

From yield curve to yield surface in wake of the financial crisis

Christian Fenger, chfe@danskebank.dk



Outline

- Yield curve building before financial crisis
- Discounting
- Super Surface



Yield curve building before financial crisis

- One term curve for all fixing estimates and discount curve
- Curve input
 - Fixings: ON, 1W, 2W, ..., 1M, 2M, 3M, 6M, ...
 - Futures (fixes to 3M fixings): JUN2011, SEP2011, DEC11, ...
 - Swaps (fixes to eg 6M fixings): 2Y, 3Y, 4Y, ..., 50Y
- Bootstrapping
- Seperate two-curve setup to handle cross currency swaps

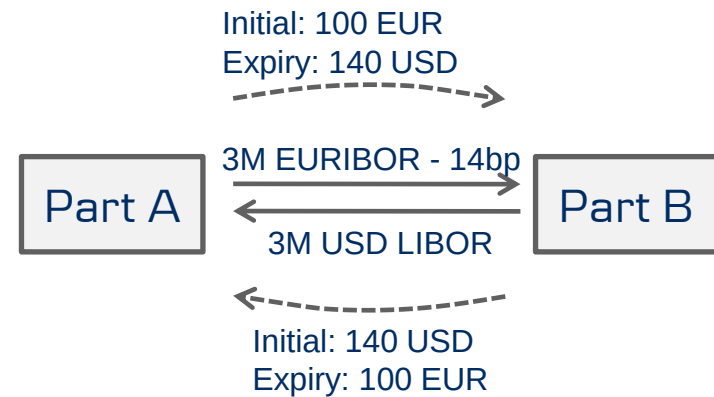
Inconsistent but okay when basis is small

Why care? Basis is large!

Tenor swap 10Y, 7mar2011



Cross currency swap 10Y, 7mar2011



Old skool yield curve building not workable



Discounting – recent developments

- Dodd-Frank Act and Basel III necessitates collateralization of swaps
 - Term sheet refers to the Credit Support Annex (CSA) to the ISDA master agreement
- Correct discounting curve is the yield on collateral
- Overnight OIS - typically

References

- Piterbarg, Funding beyond discounting: collateral agreements and derivatives pricing, Risk Magazine, Feb. 2010
- Fujii et al, Modeling of Interest Rate Term Structures under Collateralization and its Implication, SSRN, Sep 2010

Discounting when collateral

$$\begin{aligned}
 PV(t) &= E_t^Q \left[\exp\left(-\int_t^T c(s) ds\right) L(T - \Delta T) \right] \\
 &= D_c(t, T) E_t^{Q_c} [L(T - \Delta T)] \qquad F_{t,T} \equiv E_t^{Q_c^{T-\Delta T}} [L(T - \Delta T)]
 \end{aligned}$$

Proof

Assume derivative has payoff $h(T)$, and pays a dividend of $q(t)=r(t)-c(t)$

Then

$$\begin{aligned}
 dh(t) &= (r(t) - q(t))h(t)dt + dM(t) \\
 &= c(t)h(t)dt + dM(t)
 \end{aligned}$$

- c - collateral rate process
- r - risk free / funding rate
- ΔT - tenor
- M - martingale
- $L(T)$ - fixing at time T , pays at $T+\Delta T$
- $F_{t,T}$ - expected fixing at time T
- D_c - discount factors collateral rate curve

Pricing swaps with different collateral

- Assume yield surface given
- Price with another discounting curve
 - Assume collateral rate given as $c(s) + \lambda(s)$

$$\begin{aligned}
 PV &= E_t^Q \left[\exp\left(-\int_t^T c(s) + \lambda(s) ds\right) L(T - \Delta T) \right] \\
 &= D_c(t, T) E_t^{Q_c^T} \left[\exp\left(-\int_t^T \lambda(s) ds\right) L(T - \Delta T) \right] \\
 &\approx D_c(t, T) E_t^{Q_c^T} \left[\exp\left(-\int_t^T \lambda(s) ds\right) \right] E_t^{Q_c^T} [L(T - \Delta T)] \\
 &\equiv D_{c+\lambda}(t, T) E_t^{Q_c^T} [L(T - \Delta T)]
 \end{aligned}$$

c - standard collateral rate process

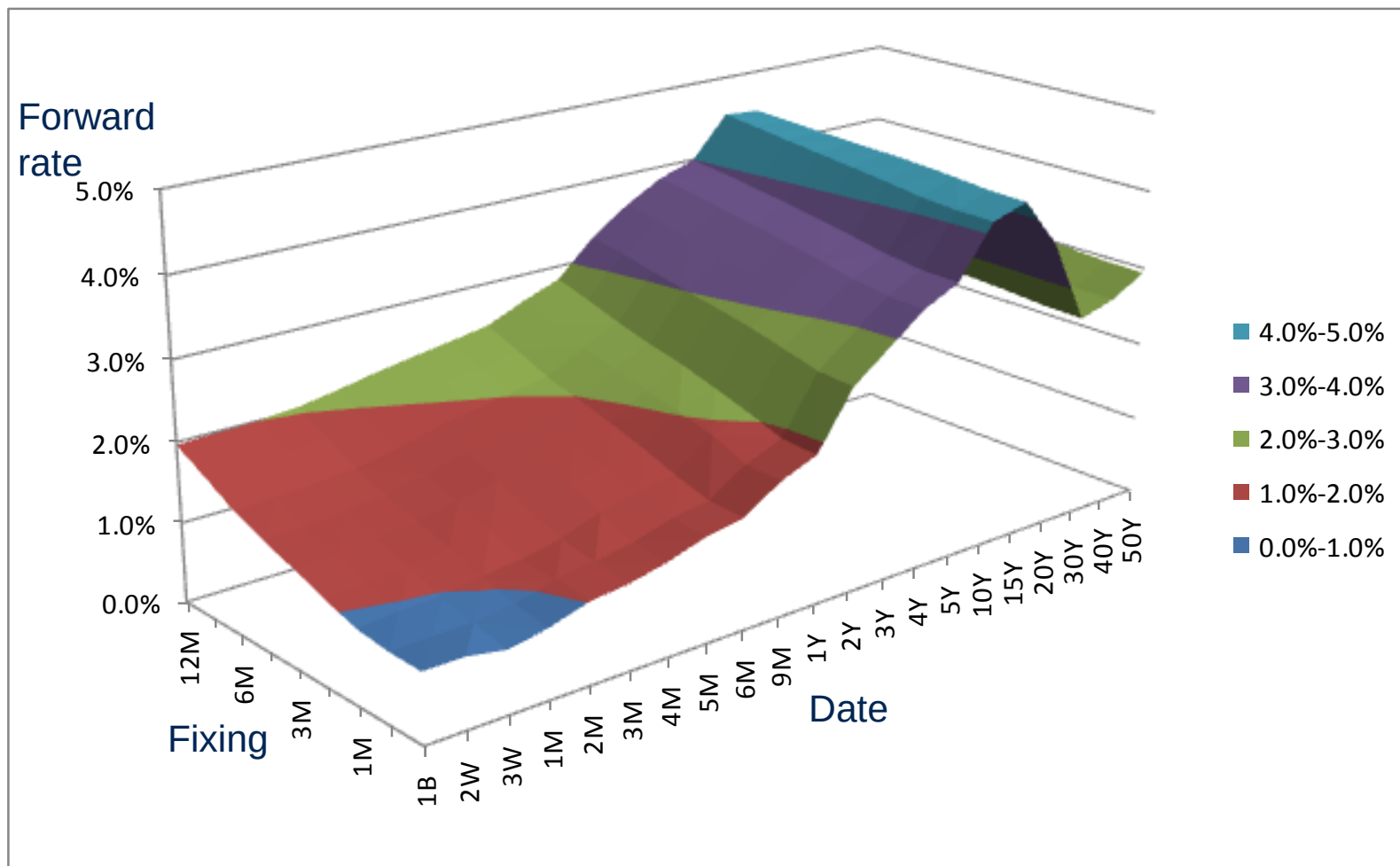
λ - spread process

$D_{c+\lambda}$ - new discount factors

Approximation valid if λ and L independent

The surface interpretation

EUR surface 7MAR2011



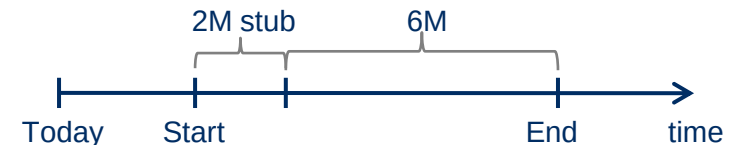
Super Surface

- One curve for each main fixing: 1M, 3M, 6M, 12M and OIS
- Turns, bumps
 - to excess or shortage of liquidity
 - Feeds into all curves
- Stubs (2D interpolation)
- Discounting correspond to the collateral agreement

Price all basis, irregular swaps, FX swaps, cross currency swaps, etc in one setup

A forward starting swap

- Trading date 7MAR2011
- Starting 15NOV2011
- End in 8M
- 8M swap against 6M fixings starting 15NOV2011
- 2M stub
 - Market does not exist
 - Interpolation in surface 2.0%
 - Interpolation in 6M curve 2.3%
 - → 9bps difference in swap rate
- Turn at New Year
 - 35bps extra in OIS rate from 30dec2011 to 2jan2012
 - → 2bps in stub rate



Challenges

2M market does not exist → Need correct interpolation

6M market illiquid in short end → Need to spread against 3M futures strip

Market input in EUR

Tenor	OIS	1M	3M	6M	12M
Very short end	FRAs	FRAs	FRAs	FRAs	FRAs
Short end	Swaps	Spread to 3M	FRA JUN11	BASIS to 3M	Spread to 6M
Long end	Spread to 3M	Spread to 3M	BASIS to 6M	Swaps	Spread to 6M

- Discounting on OIS
 - Assume collateral yields O/N OIS
- Simultaneous calibration of OIS, 3M and 6M
- Then calibration of 1M and 12M

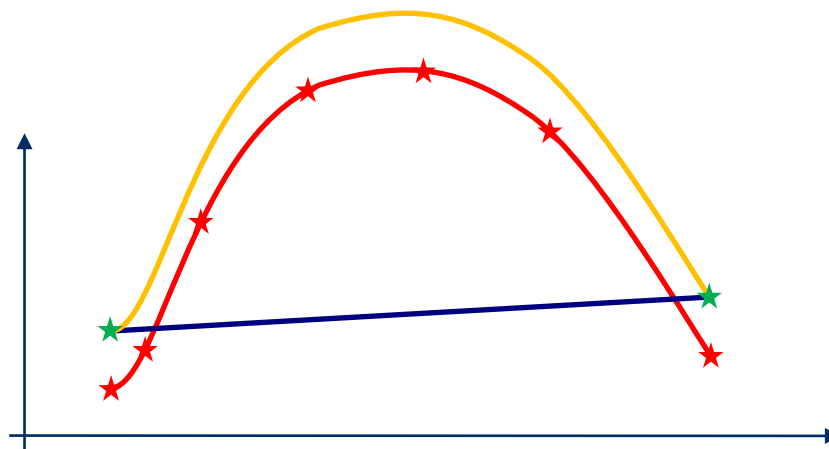


Curve building with basis in practice

- Shape
- Inheritance of shape
- Independent curve generation
- Dependent curve generation
- Splitting curves in two
 - Independent
 - Dependent
 - How to
- Real example

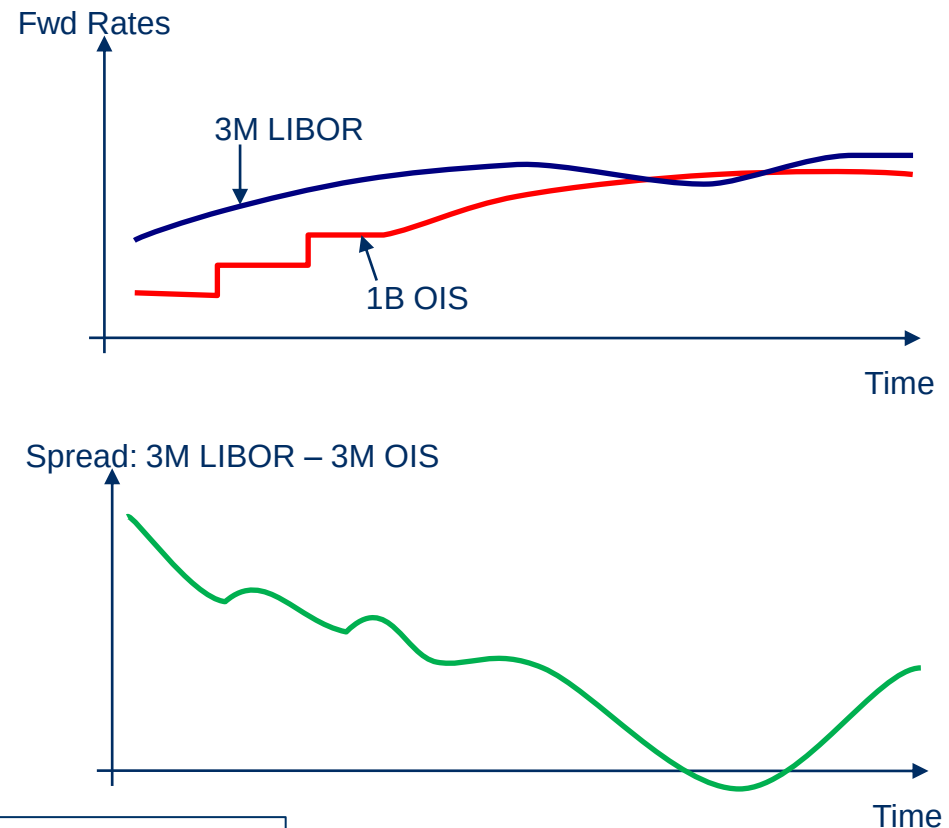
Shape of the curves

- Interpolation
 - Linear (C^0), Cubic spline (C^2), Hermite (C^1), Akima (C^1)
- ZCP verses directly in forward
- Inheritance of shape



Independent curve generation

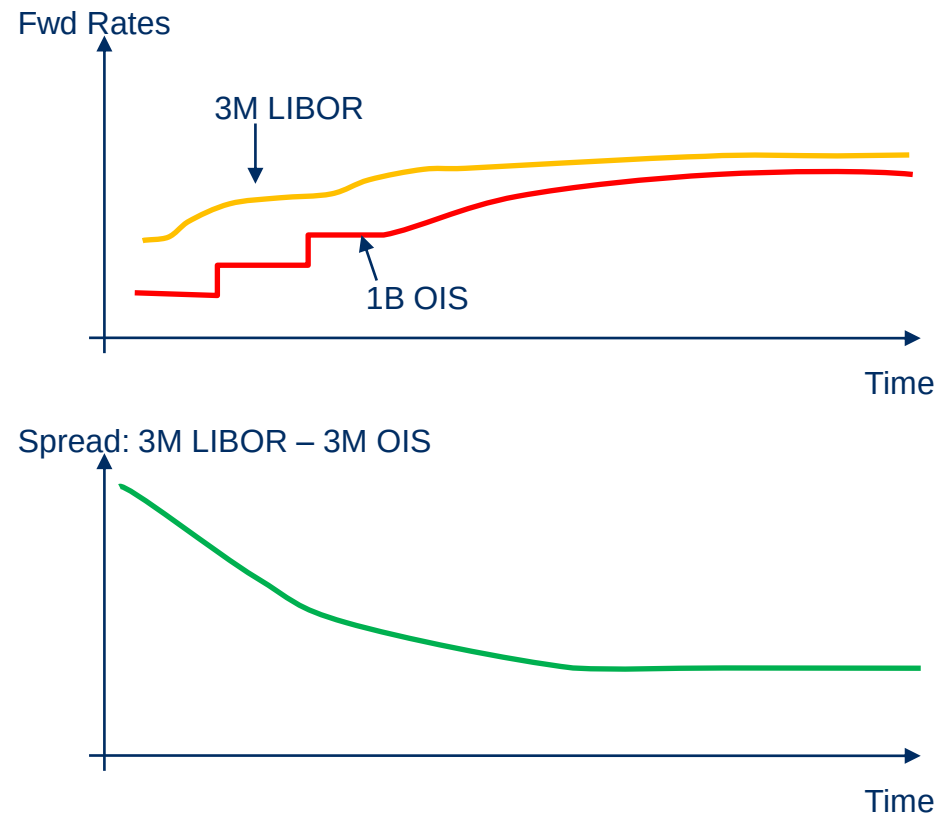
- OIS represented by one underlying curve
- 3M represented by one underlying curve



Consequence: Spread is wavy

Inheritance of shape

- OIS represented by one underlying curve
- 3M represented by a spread curve in addition to the OIS curve



Consequence: Strange interpolation between IMM FRAs

Independent and dependent simultaneously

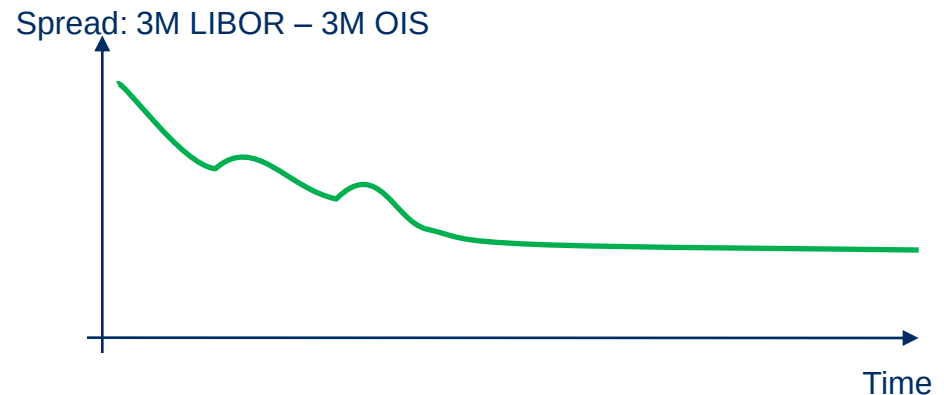
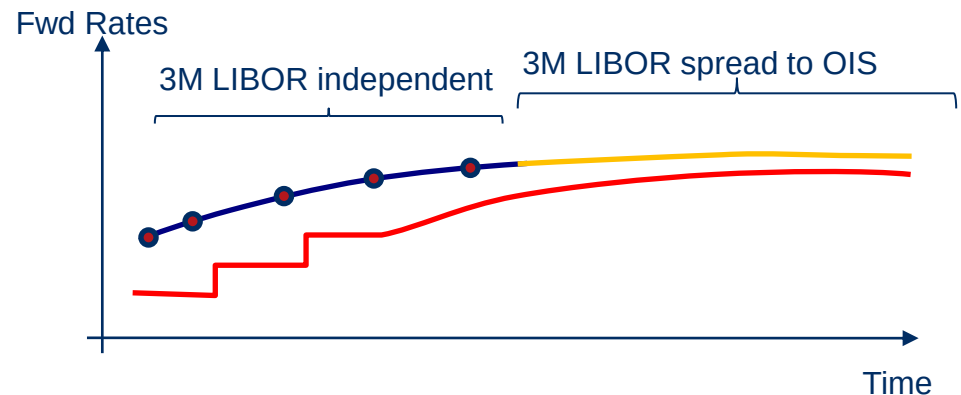
- Market observation in 3M curve

- Short end FRA/future quotes are interpolated directly
- Similar to OIS in long end

- Solution

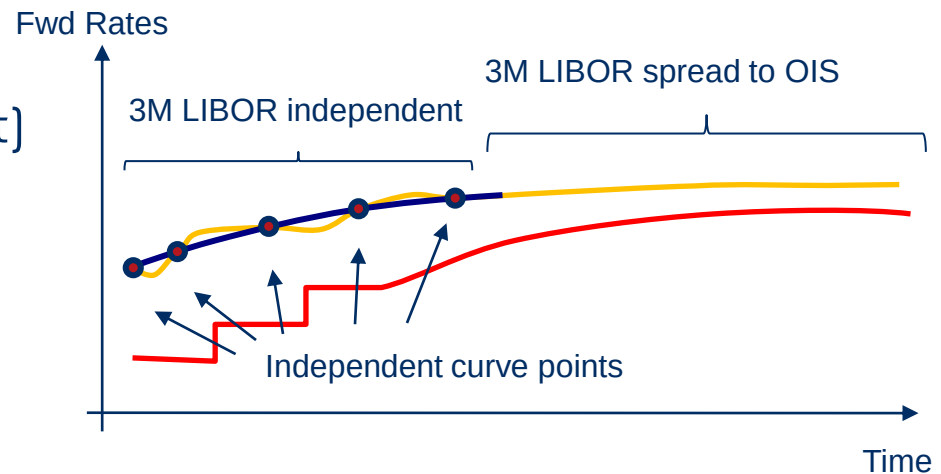
- Short end: Independent interpolation directly in forwards
- Long end: Inheritance of shape from OIS in long end

- Challenge: How to



How to Dependent / Independent

- Long end 3M rates (Inherit)
 - Look up spread
 - Look up OIS rate
- Short end 3M rates (Independent)
 - Create new curve
 - Curve points at IMM FRA starts
 - Calculate fwd rates at points
 - Look up spread
 - Look up OIS rate
 - Calculate rate from new curve with independent interpolation





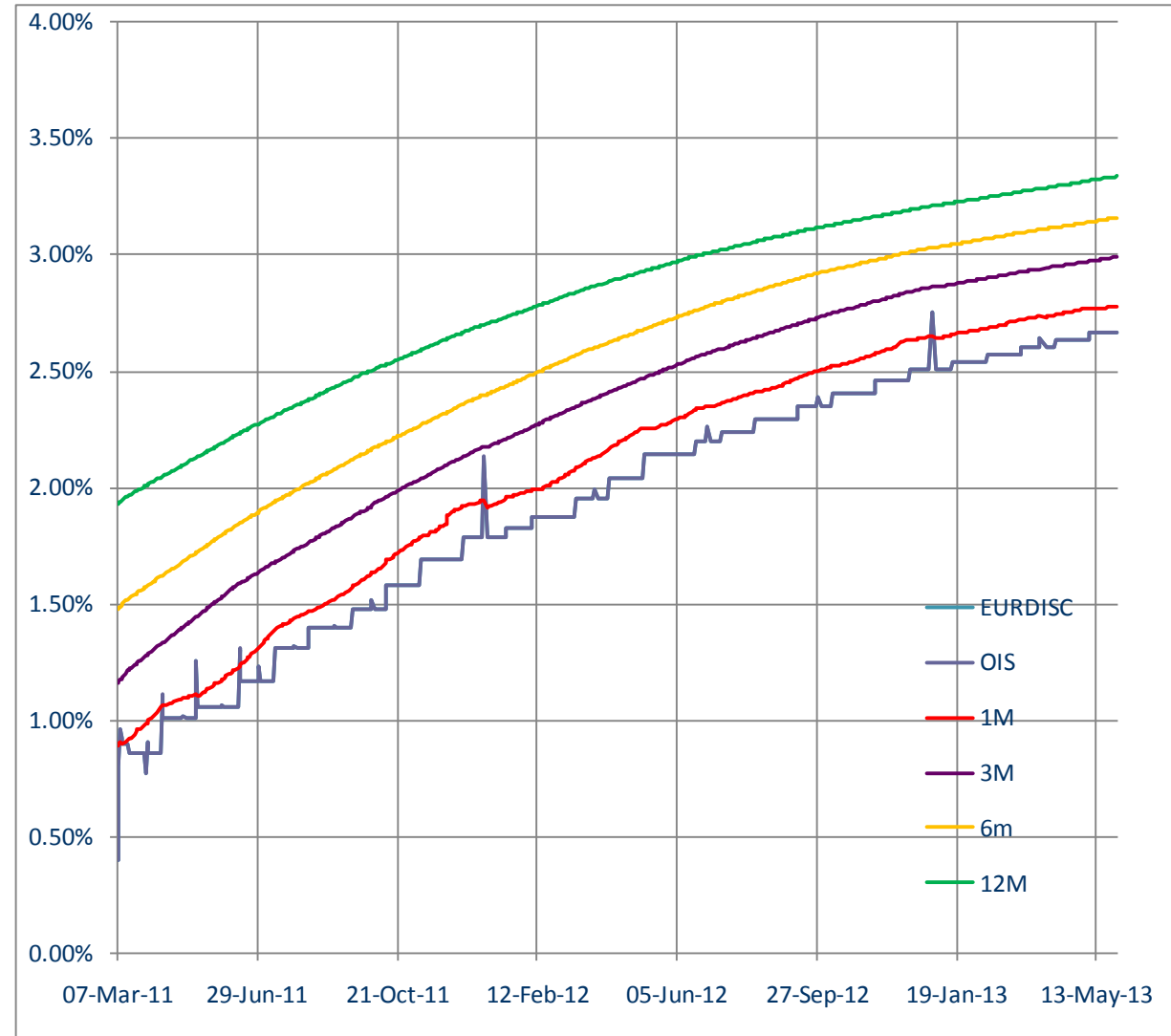
Real example: EUR surface

- OIS is main underlying curve
- Disc curve identical to OIS
- 3M inherit OIS
 - Independent in the short end
- 1M inherit OIS
- 6M inherit 3M
- 12M inherit 6M

Surface – short end

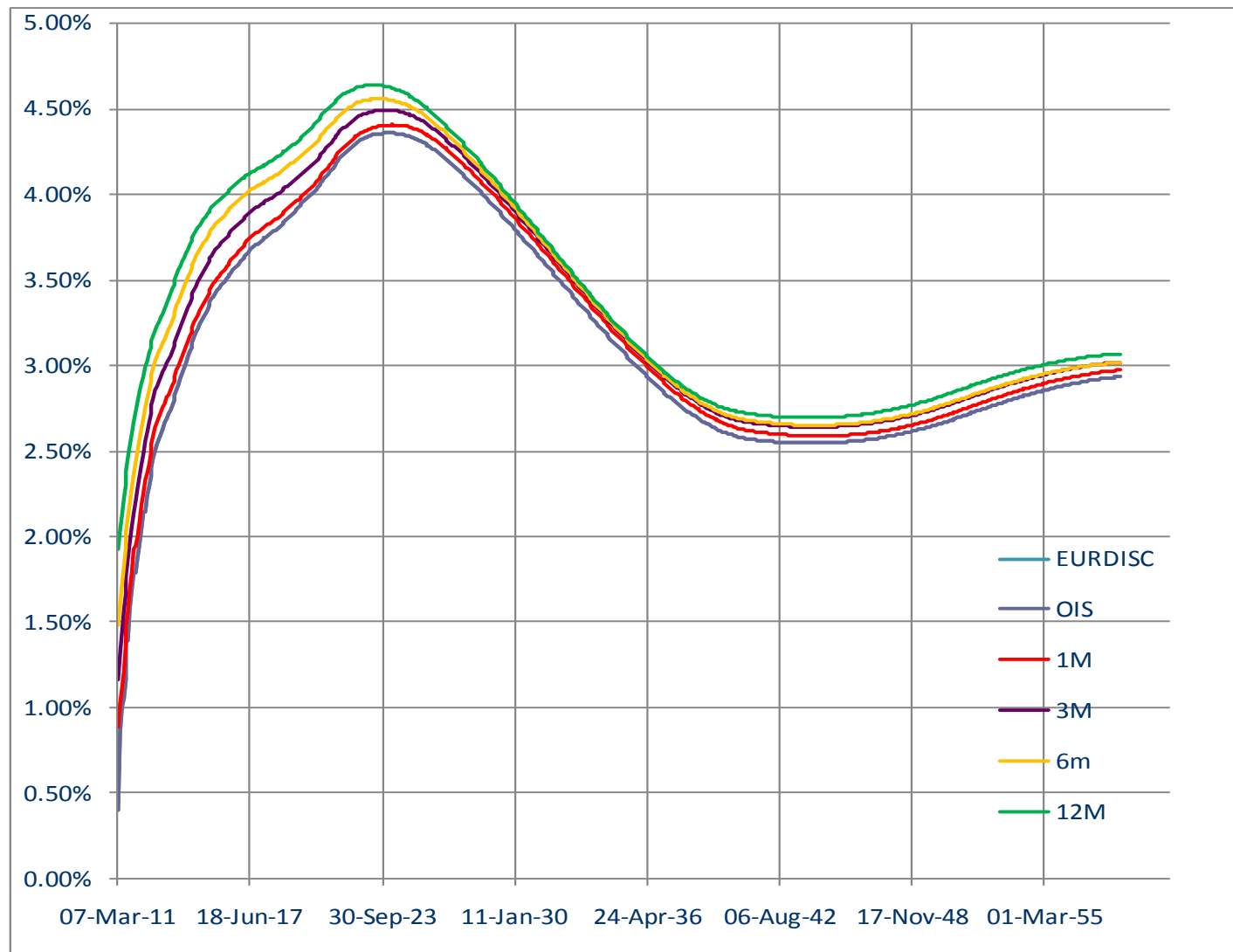
EUR 7MAR2011

- Forward curves 1M,3M,6M,12M
- OIS shown as 1B forwards
- Discount curve identical to OIS



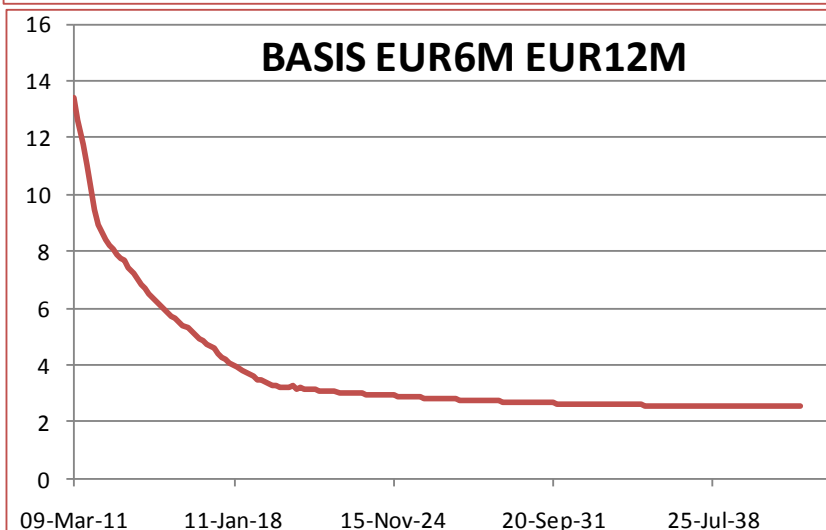
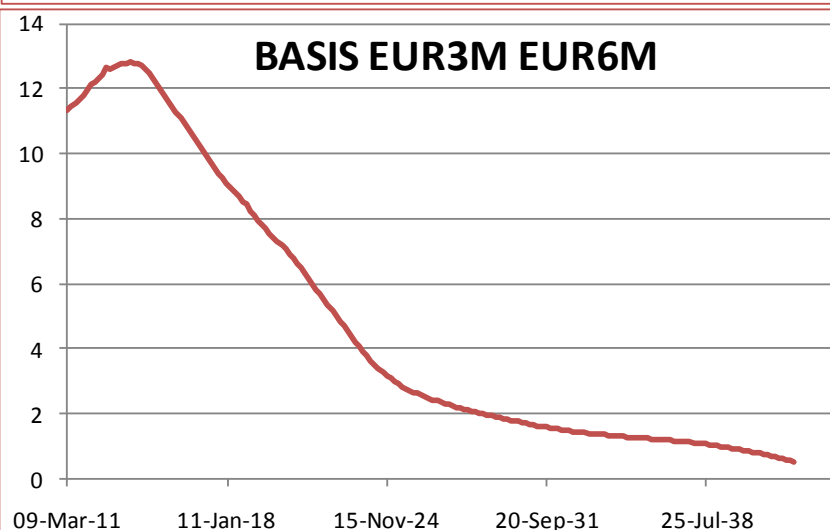
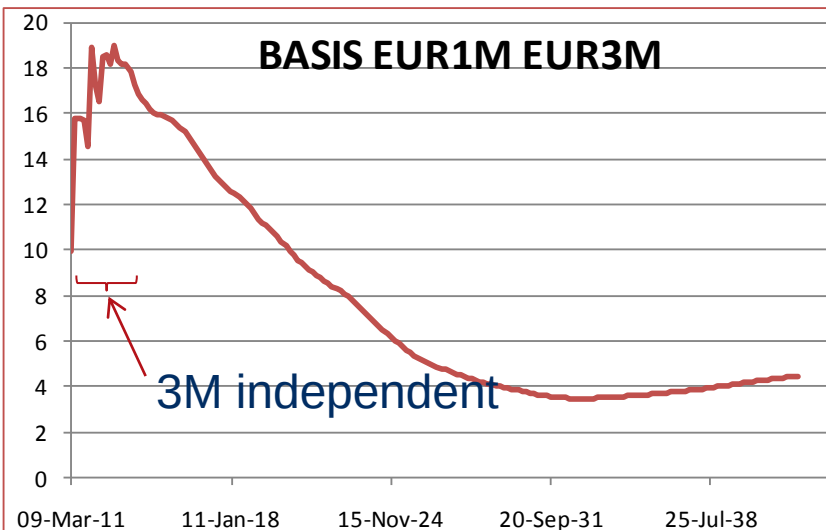
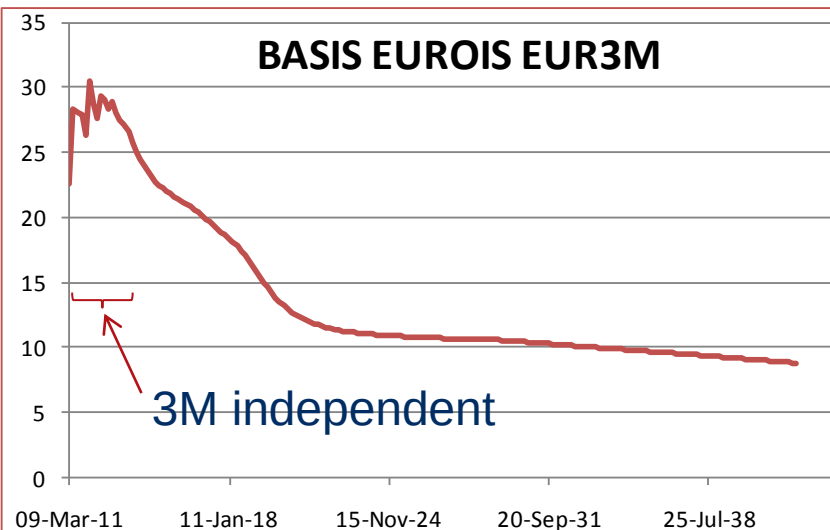
Surface – long end

EUR 7MAR2011



EUR surface – forward spreads

EUR 7MAR2011





X – currency pricing

Price instruments like

- FXFORWARD 2B 3M EURDISC USDDISC
- BASIS 3Y 1Y EUR3M USD3M

Need two surfaces



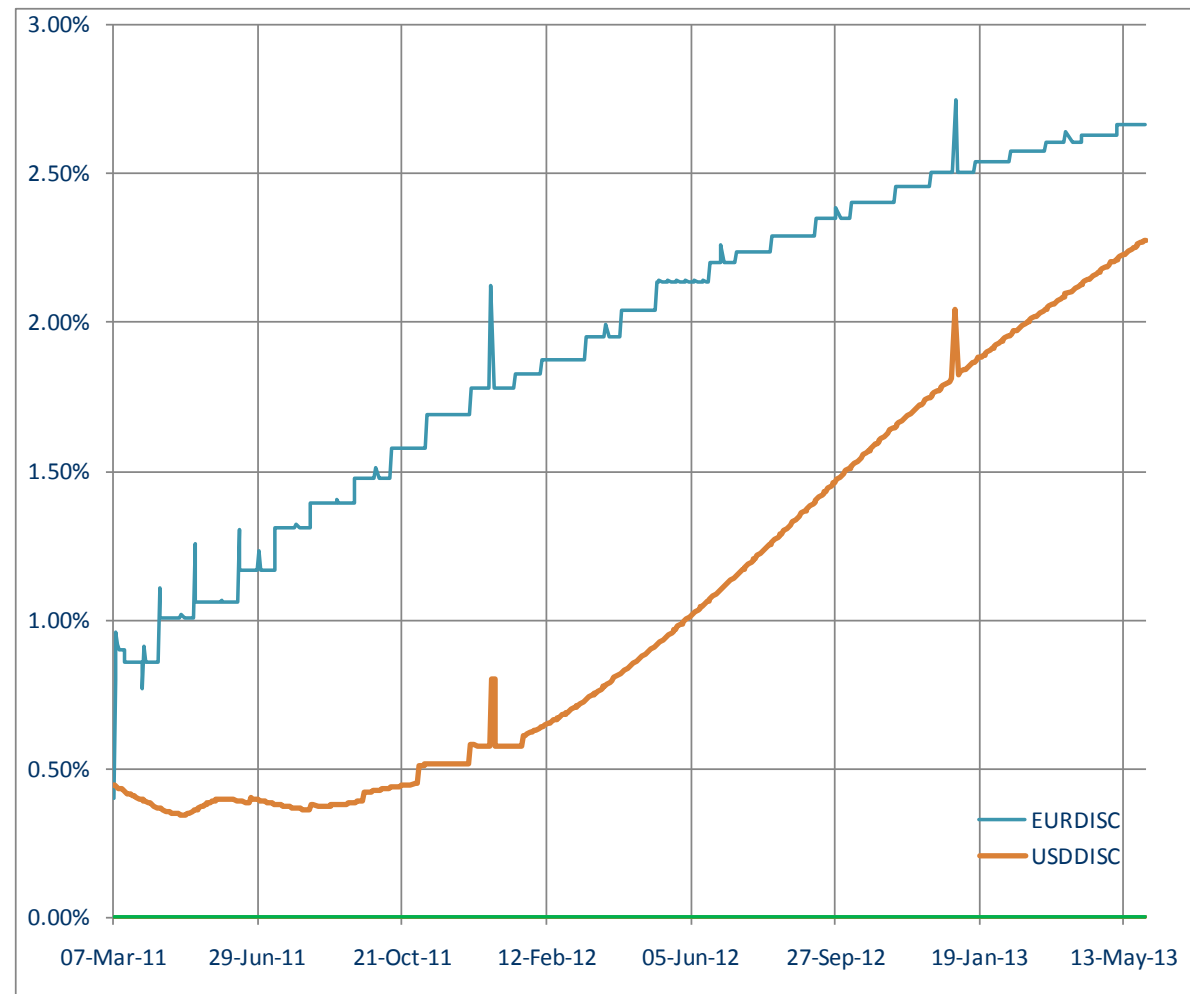
Two Surfaces

- Assume EUR given
- Calibrate USD surface
- Discounting
 - Using EUR OIS as discounting in EUR surface
 - Create new discount curve from FX forwards and cross currency swaps
 - Implicitly assuming USD instruments are collateralized with EUR OIS

Discounting – short end

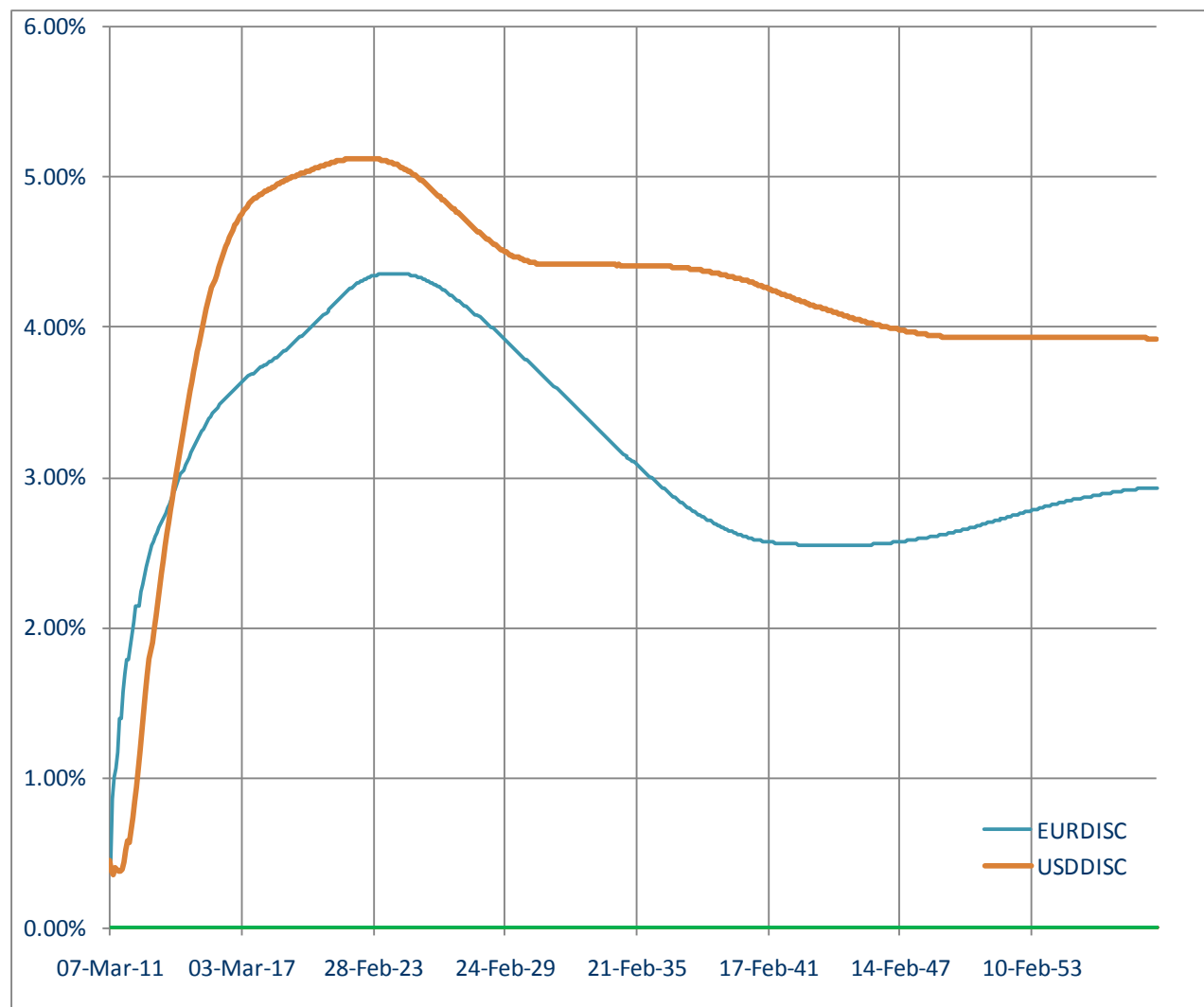
EURDISC and USDDISC, CSA EUR OIS 7MAR2011

- EURDISC foreign discount curve
- ... then level of USDDISC given
- ... form of USDDISC reflects form of cash funding



Discounting – long end

EURDISC and USDDISC, CSA EUR OIS 7MAR2011



Summary - basis in models

- Surface needed because large basis
- Choose correct discounting
- Surface
 - Inheritance of shape to get a smoother basis
 - Inheritance can be limited to long to allow for short end independence
 - Forwards curves represented directly or via ZCP
- Future: Pricing under all CSA / none CSA
 - Each surface should reflect local market collateral agreements
 - More discount curves in each surface - different instruments have different collateralization

Mathematical formulation of instruments

- FRA

$$F \leftrightarrow F_{j-1} = \left(\frac{\tilde{D}_{j-1}}{\tilde{D}_j} - 1 \right) / \delta_j$$

- SWAP

$$R \sum_{i=1}^I \delta_i \cdot D_i = \sum_{j=1}^J F_{j-1} \cdot \delta_j \cdot D_j$$

- BASIS

$$\sum_{i=1}^I (F_{i-1}^{3M} + S) \cdot \delta_i^{3M} \cdot D_i = \sum_{j=1}^J F_{j-1}^{6M} \cdot \delta_j^{6M} \cdot D_j$$

- SPREADSWAP

$$S = \frac{\sum_{j=1}^J F_{j-1}^{6M} \cdot \delta_j^{6M} \cdot D_j}{\sum_{i=1}^I \delta_i^{6M} \cdot D_i} - \frac{\sum_{k=1}^K F_{k-1}^{3M} \cdot \delta_k^{3M} \cdot D_k}{\sum_{l=1}^L \delta_l^{3M} \cdot D_l}$$

D_i - discountfactor
 $\tilde{D}_i$ - pseudo discountfactor
 F^{3M} - 3M forward rate
 S - spread
 R - fixed rate
 δ - coverage

Mathematical formulation of instruments

X_T - forward FX rate at time T
 D_T^F - foreign discountfactor
 D_T^D - domestic discountfactor
 $F^{3M,D}$ - 3M domestic forward rate
 $F^{3M,F}$ - 3M foreign forward rate
 S - spread

• FXFORWARD

$$\frac{X_T}{X_0} = \frac{D_T^F}{D_T^D}$$

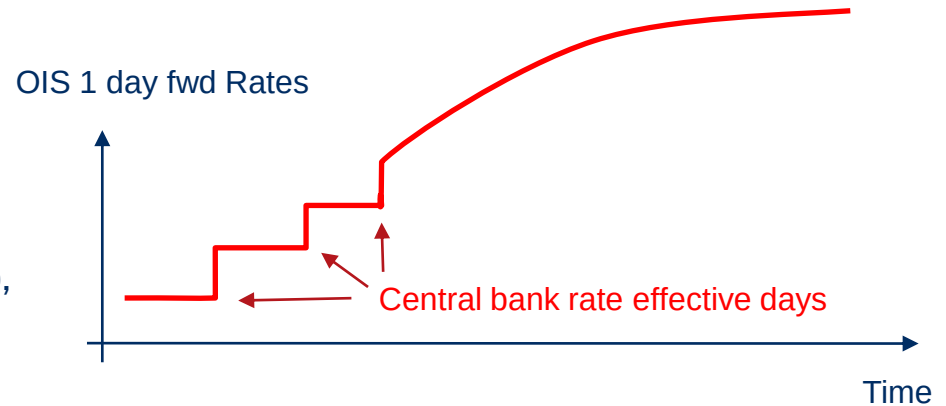
• BASIS

$$\sum_{i=1}^I F_{i-1}^{3M,D} + S \cdot \delta_i \cdot D_i^D - D_0^D + D_I^D = \sum_{j=1}^J F_{j-1}^{3M,F} \cdot \delta_j \cdot D_j^F - D_0^F + D_J^F$$

OIS curve

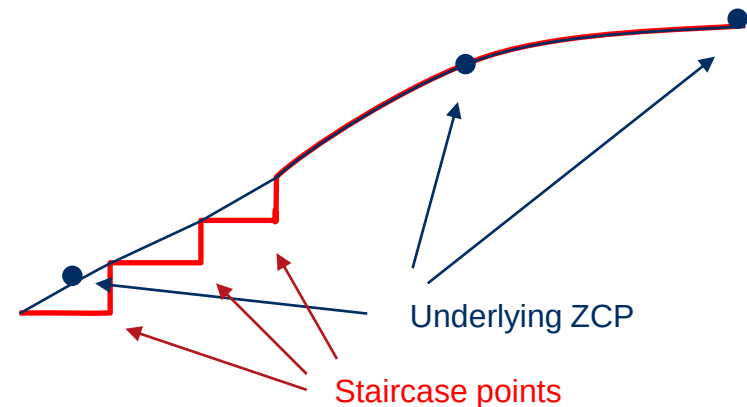
• OIS rate

- One day rate, over-night
- not a fixing but an observable rate
- expected OIS rates implied by OIS swap, Fed fund futures, etc
- directly affected by central bank
- expected OIS rates change at special central bank dates



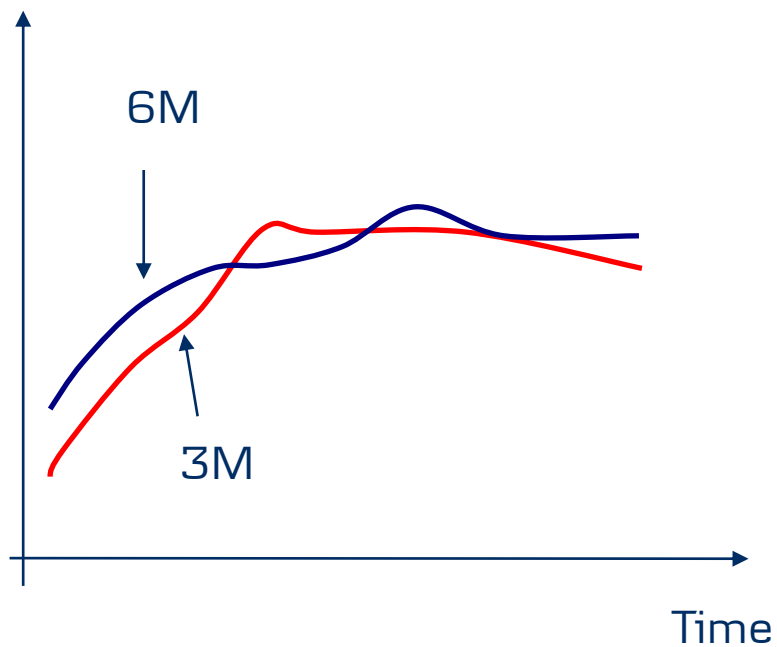
• OIS curve modelled by

- ZCP
- Staircase for 1B forwards
- reciprocal interpolation in ZCP
→ constant forward rates



Inheritance of shape

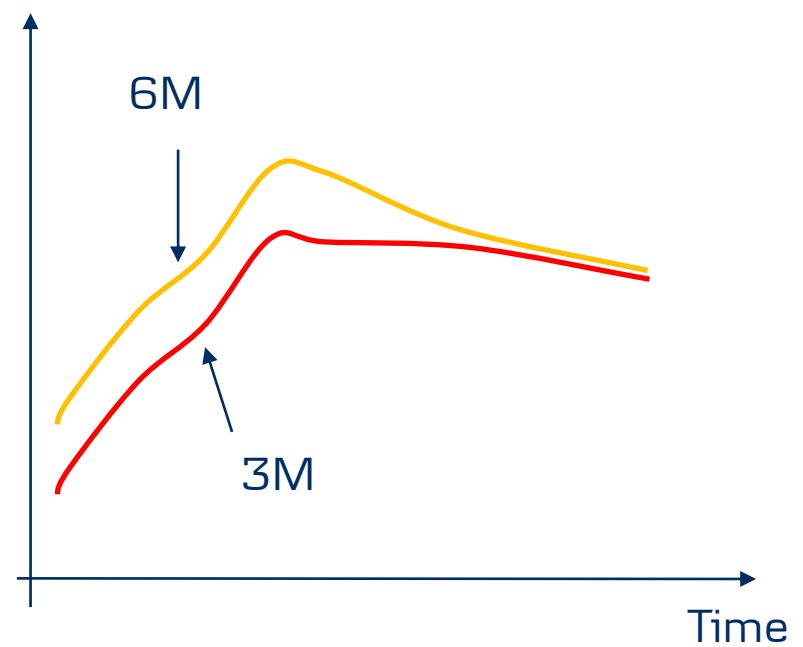
Forward Rates



Independent curve generation

- Indirect basis risk
- Difficult to build

Forward Rates



Dependent curve generation

- Direct basis risk
- Better basis PnL
- Easy to build